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Early Detection of Colorectal Cancer Following Changes to Screening Guidelines: A Population-Based Analysis of Adults Aged 45–49

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Abstract

Introduction

Colorectal cancer (CRC) is the third most common cancer worldwide, with millions of new cases and deaths annually. There has been an increasing incidence of CRC in patients less than 50 years, leading to recommendations for CRC screening initiation at age 45 in the United States by the American Cancer Society (ACS) in May 2018. However, it is unclear if CRC screening at ages 45-49 has increased detection of early-stage CRC (ES-CRC).

Method

We conducted a retrospective cohort study using Surveillance, Epidemiology, and End Results (SEER) database from 2015–2022, including individuals aged 45–49 diagnosed with CRC. The primary outcome was diagnosis of ES-CRC (in-situ or localized disease). The exposure was the time of diagnosis, categorized as pre-guideline (2015–2018) and post-guideline (2019–2022). Multivariable logistic regression assessed the association between time period and ES-CRC diagnosis, adjusting for demographic and socioeconomic factors. Subgroup analyses were conducted by sex, race, and marital status.

Results

Among 28,532 individuals, 13,413 (47.0%) were diagnosed pre-guideline and 15,119 (53.0%) post-guideline. The proportion of ES-CRC increased significantly post-guideline (31.1%) compared to pre-guideline update (29.8%, p-value = 0.012). Notably, a decline in ES-CRC diagnoses was observed in 2020, likely due to disruptions in cancer screening during the COVID-19 pandemic. The adjusted odds of ES-CRC diagnosis was higher post-guideline compared to pre-guideline update (AOR 1.068; 95% CI: 1.004–1.137). Subgroup analyses revealed higher odds of ES-CRC diagnosis among females, Asians and Black individuals post-guideline update.

Conclusion

Following the 2018 ACS guideline change, early-stage CRC diagnoses increased among adults aged 45–49. These findings suggest that the updated screening recommendations may have facilitated earlier detection in this age group.

Introduction

Colorectal cancer (CRC) is one of the most common malignancies worldwide. It is the third most diagnosed cancer and the second leading cause of cancer death globally[1]. In 2020, roughly 1.9 million new CRC cases and 0.93 million deaths occurred worldwide, with incidence rates highest in high-income regions (e.g., Australia/New Zealand and Europe)[12]. Importantly, declines in CRC incidence have occurred in older adults in many affluent countries attributed to risk-factor changes (smoking, diet, etc.)

and widespread screening[3]. However, a concerning divergence has emerged in younger populations as multiple reports document that CRC incidence among adults under 50 has been rising. For example, U.S. cancer registry data show that in 2019, 20% of CRC cases occurred in people aged ≤ 54 (up from 11% in 1995), and incidence in people aged 50 years and below grew by approximately 2% per year during the last decade[4].

Historically, most population screening guidelines recommended beginning average-risk CRC screening at age 50. However, the American Cancer Society (ACS) in 2018 lowered the recommended starting age to 45 years[5].The U.S. Preventive Services Task Force (USPSTF) followed suit in 2021, expanding its age range to 45–75 years (Grade B for ages 45–49) [6]. The rationale was to detect cancer at an earlier, more curable stage as the stage at diagnosis strongly determines outcome. The 5-year relative survival is approximately 91.5% when detected in a localized stage versus only ~ 16.2% once distant metastases are present [7]. Randomized trials and observational studies have consistently shown that screening (via colonoscopy, FIT/FOBT, or sigmoidoscopy) reduces CRC mortality by shifting diagnoses toward earlier stages (by finding precancerous polyps or early tumors)[7].

Early evidence suggests this policy shift for earlier screening should increase screening uptake and earlier detection, but its real-world impact is still unclear[8]. We therefore aim to evaluate recent epidemiologic data for people aged 45 to 49-years to determine whether the rate of early-stage CRC detection (ES-CRC) in this age group has changed since the guideline update to start screening at age 45. This analysis will help assess whether lowering the screening age is translating into measurable improvements in early diagnosis among younger adults.

Methods

Study Design and Data Source

We conducted a retrospective study using data from the Surveillance, Epidemiology, and End Results (SEER) database. The SEER database is a comprehensive source of population-based information on cancer incidence and survival in the United States. It covers approximately 48% of the U.S. population, containing patient demographics, characteristics of the tumor, and treatment received.

Study Population

The study included patients aged 45–49 who were diagnosed with colorectal cancer (CRC) between 2015 and 2022. The year 2019 was identified as an inflection point due to the update of the American Cancer Society (ACS) screening guidelines in 2018, which lowered the recommended age for CRC screening from 50 to 45 years. This allowed for a comparison of diagnoses of CRC before and after the guideline update.

Variables and Measures

The primary outcome of interest was diagnosis of early stage of CRC (ES-CRC). ES-CRC was defined as colorectal cancer in-situ and localized cancer (stage 1). The independent variable was the time period relative to the ACS guideline update with 2019 as the inflection point with pre-guideline period from 2015–2018 and post-guideline period from 2019–2022. We included multiple years before and after the inflection point to account for underlying trends, reduce short-term variability, and strengthen causal inference. Patient demographics including age at diagnosis, sex, marital status, race, median household income and urban rural continuum were obtained.

Statistical Analysis

Multivariable logistic regression analysis was used to evaluate the odds of diagnosis of ES-CRC in the specified age group before and after the guideline change adjusting for potential confounders such as sex, race, and other demographic factors. Subgroup analysis was also performed to determine the effect of the update on racial groups, sex and marital status. All analyses were performed using Stata version 18.5 (StataCorp). Data was analyzed in May 2025. A 2-sided $P < 0.05$ was considered statistically significant for all analyses.

Result

Study Population

During the study period, a total of 28,532 individuals aged 45–49 were diagnosed with colorectal cancer (CRC) and included in the analysis. Of these, 13,413 individuals (47.0%) were diagnosed before the screening guidelines update, while 15,119 individuals (53.0%) were diagnosed after the update. Majority of the population were male, married, White population, had median household income of $60,000-$79,000 and lived in counties in metropolitan areas with more than 1 million population.

Distribution of Early-Stage CRC Diagnoses

Among the total CRC diagnoses, 8,695 cases (30.5%) were identified as early-stage colorectal cancer. There was a statistically significant increase in the proportion of ES-CRC diagnoses post-guideline update (4,705 cases, 31.1%) compared to the pre-update period (3,991 cases, 29.8%, $p = 0.012$), suggesting an improvement in early detection following the change in screening recommendations (Table 1). Notably, as illustrated in Fig. 1, there was a decline in the proportion of ES-CRC diagnoses in 2020, likely influenced by the COVID-19 pandemic, which disrupted routine healthcare services, including cancer screening.

Table 1
Baseline Characteristics of Study Population

	Pre-Screening Update (2015–2018)	Post-Screening Update (2019–2022)	p-value
CRC Diagnosis	N = 13,413	N = 15,119	
Early-Stage CRC Diagnosis	3,991 (29.8%)	4,705 (31.1%)	0.012
Year of diagnosis			
2015	3,281 (24.5%)		< 0.001
2016	3,262 (24.3%)		
2017	3,478 (25.9%)		
2018	3,392 (25.3%)		
2019		3,508 (23.2%)	
2020		3,333 (22.0%)	
2021		3,786 (25.0%)	
2022		4,492 (29.7%)	
Sex			
Female	6,056 (45.2%)	6,924 (45.8%)	0.27
Male	7,357 (54.8%)	8,195 (54.2%)	
Marital Status			
Not married	3,674 (40.6%)	3,974 (38.7%)	0.007
Married	5,386 (59.4%)	6,306 (61.3%)	
Race			
White persons	7,464 (56.0%)	7,705 (51.6%)	< 0.001
Black persons	1,831 (13.7%)	2,159 (14.5%)	
Hispanic persons	2,764 (20.7%)	3,477 (23.3%)	
Asian persons	1,149 (8.6%)	1,469 (9.8%)	
American Indian or Alaska Native persons	113 (0.8%)	108 (0.7%)	
Median Household Income			
<$40,000-$59,000	1,754 (13.1%)	1,625 (10.7%)	<

	Pre-Screening Update (2015–2018)	Post-Screening Update (2019–2022)	p-value
			0.001
$60,000-$79,000	5,351 (39.9%)	4,381 (29.0%)	
$80,000-$99,000	3,253 (24.3%)	5,010 (33.1%)	
$100,000 and above	3,055 (22.8%)	4,103 (27.1%)	
Rural Urban Continuum			
Counties in metropolitan areas with > 1 million population	8,734 (65.3%)	9,996 (66.2%)	< 0.001
Counties in metropolitan areas with 250,000–1 million population.	2,265 (16.9%)	2,704 (17.9%)	
Counties in metropolitan areas with < 250,000 population	898 (6.7%)	963 (6.4%)	
Nonmetropolitan counties adjacent to a metropolitan area	955 (7.1%)	976 (6.5%)	
Nonmetropolitan counties not adjacent to a metropolitan area	530 (4.0%)	467 (3.1%)	

Statistical Analysis and Findings

The adjusted odds ratio (AOR) for ES-CRC diagnosis post-guideline change was 6.8% higher compared to ES-CRC diagnosis pre-guideline update (AOR: 1.068 95% CI: 1.004–1.137, p-value = 0.038). Subgroup analysis showed that females (AOR:1.104, 95% CI: 1.009–1.208 p -value = 0.032), Asians (AOR:1.388, 95% CI:1.137–1.694 p-value = 0.001) and Black populations (AOR:1.241, 95% CI:1.051–1.466 p-value = 0.011) had significantly higher odds of ES-CRC diagnosis after the screening guidelines update as shown in Fig. 2.

Discussion

In this retrospective study using SEER data, 28,532 individuals aged 45–49 diagnosed with CRC from 2015 to 2022 were analyzed. Following the ACS guideline change, the proportion of early-stage CRC diagnoses modestly but significantly increased from 29.8–31.1% (p = 0.012). Multivariable logistic regression confirmed a 6.8% higher adjusted odds of early-stage diagnosis post-guideline change. This modest improvement is in line with prior predictive modeling that suggested that lowering the screening age would lead to stage migration and increased early detection rates leading to increased survival[9,10].However, national data show that CRC screening uptake remains low in this age group. According to the 2021 National Health Interview Survey, only about 20% of adults aged 45–49 years were up to date with CRC screening, the lowest rate among eligible age groups[4]. As awareness of the

guideline increases and uptake improves over time, the impact of earlier screening on detection rates may become more pronounced.

In the subgroup analysis, females demonstrated significantly greater odds of early-stage colorectal cancer (CRC) detection following the implementation of new screening guidelines (OR: 1.104, 95% CI: 1.009–1.208, $p = 0.032$), whereas the increase among males was not statistically significant. This sex-specific difference may be partly explained by variation in healthcare utilization and preventive care behavior. National CDC data indicate that, even excluding pregnancy-related visits, women are 33% more likely than men to routinely seek physician care, creating more frequent opportunities for screening and early detection[11]. Additionally, multivariable analysis from a study of Black Medicare beneficiaries in Baltimore found that men had 28% lower odds of being up to date with CRC screening compared to women (OR: 0.72, 95% CI: 0.52–0.99)[12]. These patterns point to a persistent behavioral and access gap that may contribute to earlier stage diagnosis in women and reinforce the need for targeted strategies to improve screening adherence among men. Similarly, Black individuals (AOR = 1.241, p = 0.011) were more likely to be diagnosed with early-stage CRC following the guideline change. This observation is consistent with previous literature showing that minority ethnic groups especially Black Americans bear a disproportionately high burden of CRC incidence and mortality[13]. However, studies also show that when equitable access to high-quality screening is provided, Black populations experience substantial benefits. For example, Alagoz et al. demonstrated that eliminating disparities in follow-up colonoscopy rates and endoscopist detection quality could reduce CRC incidence and mortality by 14.6% and 18.7%, respectively, and increase life-years gained by 7.1%, ultimately closing up to 59% of the CRC mortality gap between Black and White adults[14].Although non-Hispanic Black adults have long been recognized as a high-risk group, with the American College of Gastroenterology recommending earlier screening starting at age 45 years as far back as 2005, these race-specific recommendations were inconsistently implemented and lacked broad support from payer reimbursement policies[15]. Consequently, this had a limited impact on population-level outcomes. Notably, our analysis compares time periods entirely after this 2005 recommendation, and the observed increase in early-stage CRC diagnoses among Black individuals became evident following the screening guideline update. This suggests that the standardized, universally endorsed screening guideline update likely played a more direct role in improving access and early detection in this population.

Asian individuals also showed notably higher odds of early-stage CRC detection following the guideline change (AOR = 1.388, p = 0.001). While the underlying factors are not fully understood, this may reflect regional differences in healthcare access, screening uptake, or cultural attitudes toward preventive care. Further investigation is needed to explore these trends.

Impact of COVID 19 on screening

A notable dip in early-stage diagnoses in 2020 highlights the disruption caused by the COVID-19 pandemic, which significantly reduced access to preventive services[16].Colonoscopy volumes reportedly

dropped by more than 70% during the early months of the pandemic however there was a resurgence to pre pandemic levels after[17].

Limitations

This study has limitations. First, its observational design limits the ability to draw causal conclusions, and the possibility of residual confounding from unmeasured factors such as family history, comorbidities, or access to care cannot be ruled out. Secondly, although the ACS guideline change occurred in 2018, real-world screening uptake may have been gradual. As such, some post-2019 diagnoses may still reflect cancers identified outside formal screening programs. Moreover, while the ACS recommended starting screening at age 45 in 2018, most insurers did not adopt this change until the USPSTF issued similar guidance in 2021. Because insurance coverage is typically aligned with USPSTF recommendations, early uptake in the 45–49 age group may have been limited, potentially underestimating the full effect of the guideline update during the study period.

Conclusion

Our study suggests that the American Cancer Society update to CRC screening guidelines, which lowered the recommended starting age to 45, may be contributing to a modest but meaningful increase in early-stage colorectal cancer diagnoses among U.S. adults aged 45–49. While the absolute gain is small, the trend aligns with expectations from modeling studies and highlights the potential value of earlier screening in improving outcomes. Notably, certain subgroups particularly women, Black individuals, and Asian individuals appear to have derived greater benefit, emphasizing the importance of equitable access and culturally informed outreach strategies. However, given the observational nature of the study and lack of screening-specific data, further prospective research is needed to confirm these patterns and better understand the mechanisms driving them. Continued efforts to monitor screening uptake, address disparities, and support timely follow-up remain essential to maximizing the impact of guideline changes on population health.

Declarations

Financial disclosure

None to disclose

Author Contribution

B.M: conception, methodology, analysis, writing – original draft preparation, reviewing and editing. A.O: literature search, writing – original draft preparation. G.G: literature search, writing – reviewing and editing. N.A: literature search, writing – original draft preparation. S.H: writing– original draft preparation.

R.B: writing – reviewing and editing. R.V: writing – reviewing and editing. J.A: writing – reviewing and editing. S.S: writing – reviewing and editing. P.K: supervision, writing- reviewing and editing.

Data Availability

Data used in this study can be accessed from the Surveillance, Epidemiology, and End Results Program database via the link https://seer.cancer.gov/data/

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Figures

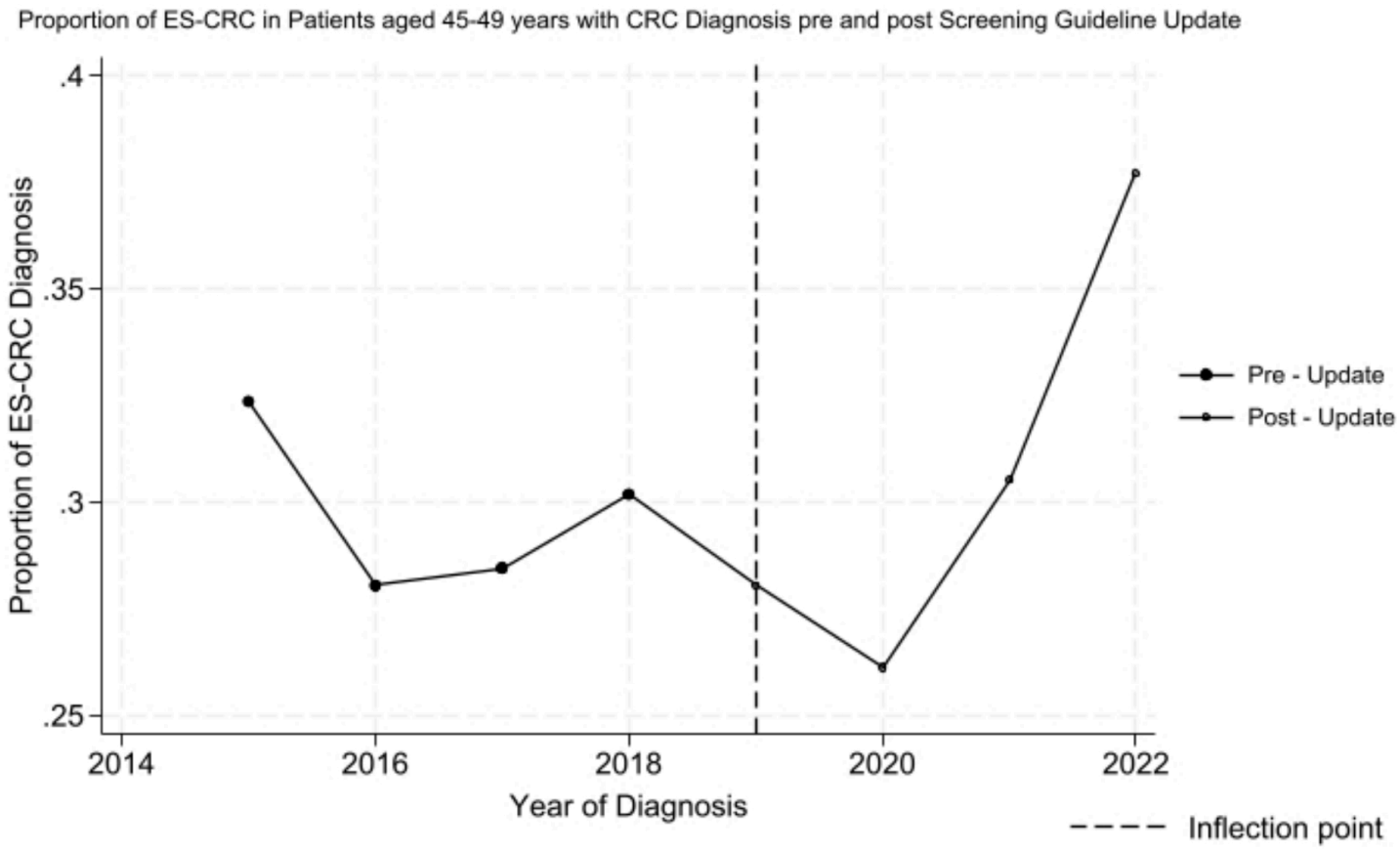


Figure 1

Proportion of Early CRC Diagnoses Per Year

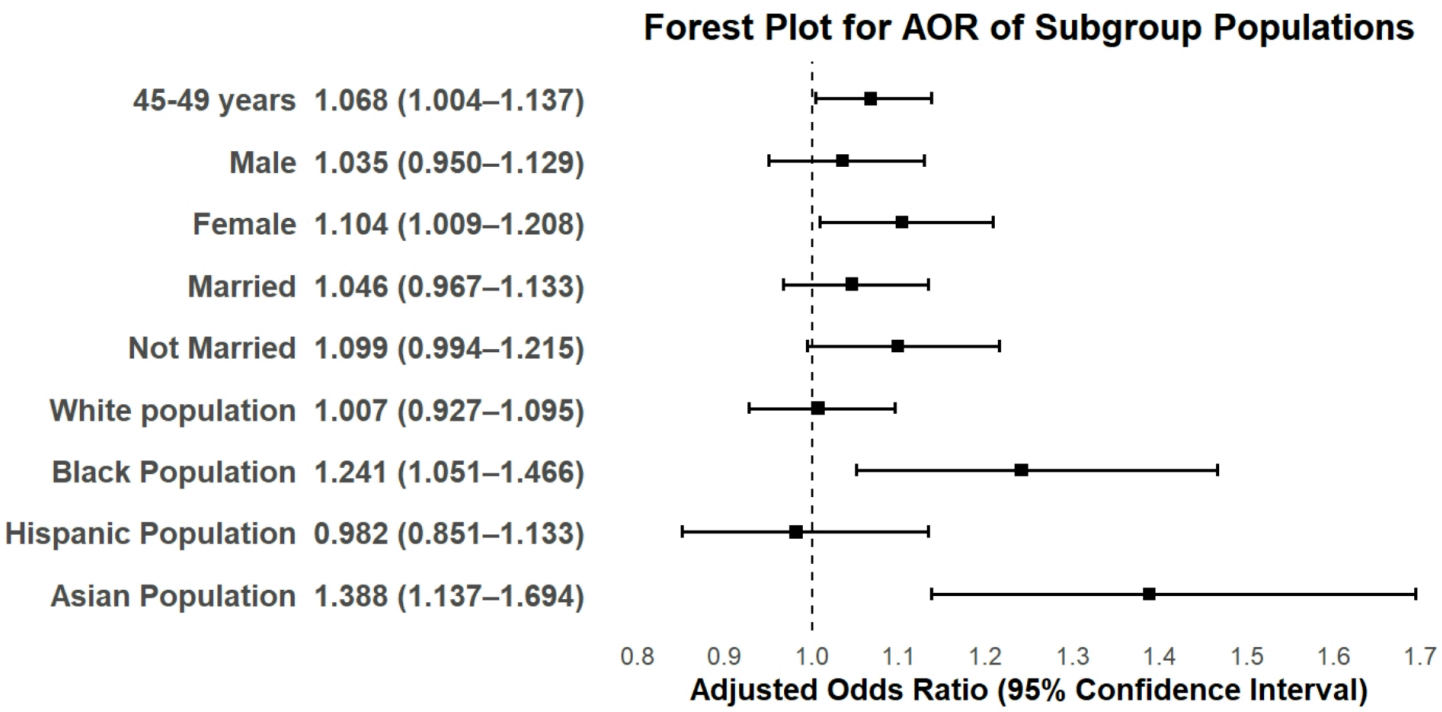


Figure 2

Forest Plot of Adjusted Odd Ratios of ES-CRC diagnosis in Subgroup Populations